IMPULSE AND MOMENTUM

The differential equation of rectilinear motion is

$$\frac{W}{g}\ddot{x} = \Sigma F_x$$

$$\frac{W}{g}\frac{d\dot{x}}{dt} = \Sigma F_x$$

$$d\left(\frac{W}{g}\dot{x}\right) = \Sigma F_x . dt$$

The quantity $\left(\frac{W}{g}\dot{x}\right)$ is known as the momentum of the particle and $(\Sigma F_x . dt)$ is known as the impulse of the external force ΣF_x in time dt. The above expression says that the differential change in the momentum of the particle in time dt is equal to the impulse of the force during that time. Integrating both sides knowing that when $t = 0, \dot{x} = \dot{x}_0$

$$\frac{W}{g}\int_{\dot{x}_0}^{\dot{x}} d\dot{x} = \int_0^t \Sigma F_x . dt$$

$$\frac{W}{g}\dot{x} - \frac{W}{g}\dot{x}_0 = \int_0^t \Sigma F_x . dt$$

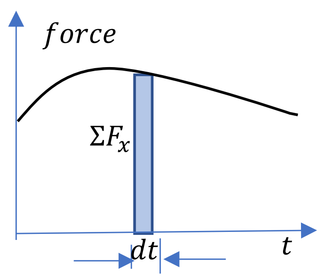
This states that the change in momentum of the particle during a time interval is equal to the impulse of the external force during the same time. Or in other words,

$$\frac{W}{g}\dot{x}_0 + \int_0^t \Sigma F_x . dt = \frac{W}{g}\dot{x}$$

Initial momentum + Impulse of the force = Final momentum

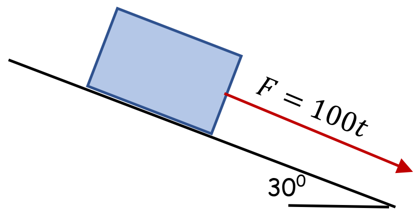
This is known as Impulse-Momentum principle. Both impulse and momentum are vector quantities having directions same as that of the force and the velocity respectively.

We can evaluate $\int_0^t \Sigma F_x . dt$ only if ΣF_x is a constant or a function of time. Suppose ΣF_x is a function of time. Then the graph shows the force-time diagram. Let ΣF_x be the magnitude of force during the infinitesimally small time dt. Then $\Sigma F_x . dt$ represents the area of the infinitesimal area of the force-time diagram. Therefore, $\int_0^t \Sigma F_x . dt$ represents the total area under the force-time diagram from time 0 to t.



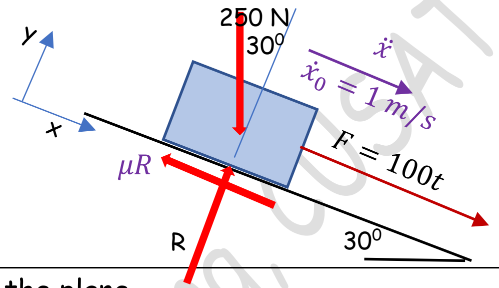
Example 1

A body having a weight of $250\ N$ is acted upon by a force $F = 100\ t\ N$, where t is in seconds. Determine the body's velocity $2\ s$ after the force has been applied. The body has an initial velocity $\dot{x}_0 = 1\ m/s$ down the 30^0 inclined plane and the coefficient of kinetic friction between the body and the plane is 0.3.



Draw the free-body diagram. The forces are

1. Applied force $F = 100\ t\ N$
2. Weight $250\ N$
3. Normal reaction R
4. Frictional force μR



Resolving the forces along the plane and perpendicular to the plane.

$$\Sigma F_y = 0 \quad \ggg \quad R - 250\cos 30 = 0 \quad \ggg \quad R = 216.5\ N$$

Since the motion is along the plane, let us write the impulse-momentum principle along the plane.

Initial momentum of the body along the plane,

$$\frac{W}{g}\dot{x}_0 = \frac{250}{9.81} \times 1 = 25.48\ kg.m/s$$

Impulse of the forces along the plane,

$$\int_0^t \Sigma F_x.dt = \int_0^2 (100\ t + 250\sin 30 - 0.3 \times 216.5)\ dt = 200 + 250 - 129.9 = 320.1\ kg.m/s$$

Final momentum of the body along the plane = Initial momentum + Impulse $= 345.58\ kg.m/s$

$$\frac{W}{g}\dot{x} = 345.58$$

$$\dot{x} = 13.56\ m/s$$

The problem can also be solved by Newton's law/DÁlembert's principle

The equation of motion along the plane

$$\frac{W}{g}\ddot{x} = \Sigma F_x \quad \ggg \quad \frac{W}{g}\ddot{x} = 100\ t + 250\sin 30 - 0.3 \times 216.5 = 100t + 60.5$$

$$\ddot{x} = \frac{9.81}{250}(100\ t + 60.5) = 3.924\ t + 2.374$$

The acceleration is a function of time.

$$\frac{d\dot{x}}{dt} = 3.924\ t + 2.374 \quad \ggg\gg \quad d\dot{x} = (3.924\ t + 2.374)\ dt$$

Integrating both sides with respect to time knowing that when $t = 0$, $\dot{x}_0 = 1\ m/s$

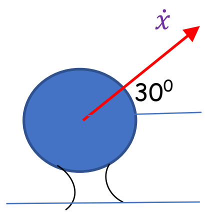
$$\int_1^{\dot{x}} d\dot{x} = \int_0^2 (3.924\ t + 2.374)\ dt$$

$$\dot{x} - 1 = 3.924\left(\frac{t^2}{2}\right)_0^2 + 2.374(t)_0^2 = 12.5$$

$$\dot{x} = 13.5\ m/s$$

Example 2

A $40\ g$ golf ball is struck in $3\ ms$ by a player such that it is given a velocity $\dot{x} = 35\ m/s$ directed to the right 30^0 from the horizontal. Determine the average impulsive force exerted on the ball. Neglect the impulse caused by the ball's weight while it is struck.



The golf ball which is initially at rest is struck by the player to give it a velocity $\dot{x} = 35\ m/s$

Since the golf ball is at rest initially, the initial momentum $= 0$

After it is struck, it acquires a velocity $\dot{x} = 35\ m/s$

Final momentum of the ball $= \frac{W}{g}\ \dot{x} = 40 \times 10^{-3} \times 35 = 1.4\ \ kg.m/s$ (in the direction of velocity $\dot{x}$)

By impulse-momentum principle, final momentum=initial momentum + impulse

$\therefore$ Impulse of the force $= 1.4\ \ kg.m/s$ (N.s)

Average impulsive force $\times$ time during which it acts $= 1.4\ \ N.s$

$\therefore$ The average impulsive force$= \frac{1.4\ N.s}{3\times10^{-3} s} = 467\ N$ inclined at 30^0 to the horizontal

IMPULSE-MOMENTUM PRINCIPLE FOR A SYSTEM OF PARTICLES

When we consider a system of particles, the internal forces always occur as equal and opposite pairs. Hence the impulses produced by these internal forces are also equal and opposite. They cancel each other. Therefore, in a system of particles if there are no external forces acting, the net impulse of the forces is zero. Then, the final momentum of the system will be equal to the initial momentum of the system. Or in other words, the momentum of the system remains conserved. This is known as the principle of conservation of momentum.

Example 1

A car A of weight $22\ kN$ is travelling to the right at $1\ m/s$. Meanwhile a $15\ kN$ car B is travelling to the left at $2\ m/s$. If the cars crash head on and become entangled, determine their common velocity just after collision and the impulse. Assume that the brakes are not applied during motion.

Here we have two cars constituting the system and the velocity is taken as positive towards right.

Initial momentum of car A $= \frac{22\times10^3}{9.81} \times 1 = 2242.61\ kg.m/s$

Initial momentum of car B $= -\frac{15\times10^3}{9.81} \times 2 = -3058.1\ kg.m/s$ (negative because velocity is towards left)

Initial momentum of the system$= 2242.61 - 3058.1 = -815.49\ kg.m/s$

Let the vehicles move together after collision with velocity $\dot{x}$

Final momentum of the system$= \frac{(22+15)10^3}{9.81}\ \dot{x} = 3771.66\dot{x}$

Since no external forces act on the system during collision, the momentum of the system is conserved.

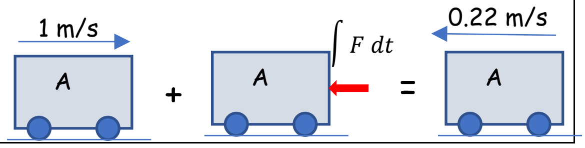
∴ Initial momentum of the system=final momentum

$$3771.66\dot{x} = -815.49$$

$$\dot{x} = -0.22\ m/s$$

After collision, the entangled vehicles move towards left with velocity $0.22\ m/s$.

To find the impulse, let us consider any one car.



Initially the car A is moving towards right with velocity $1\ m/s$. During collision, the impulse $\int F\,dt$ of the force F exerted by car B act on the car A. After collision, the car A moves towards left with velocity $0.22\ m/s$

Initial momentum of car A $= 2242.61\ kg.m/s$ (towards right)

Final momentum of car A $= -\frac{22\times10^3}{9.81} \times 0.22 = -493.37\ kg.m/s$ (towards left)

According to Impulse-Momentum Principle, Initial momentum + Impulse=Final momentum

Initial momentum $- \int F\,dt$ =Final momentum

$\int F\,dt = 2242.61\ kg.\frac{m}{s} + 493.37\ kg.\frac{m}{s} = 2735.98\ N.s$

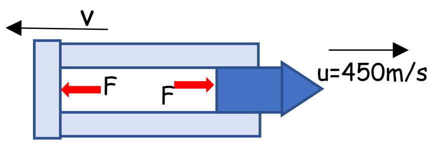
Impulse= $2735.98\ N.s$ (towards left)

Example 2

A $6000\ N$ gun fires a $40\ N$ shell with a muzzle velocity of $450\ m/s$ relative to the ground. If the firing takes place in $0.03\ s$, determine the recoil velocity of the gun after firing. Also find the average impulsive force acting on the shell.

The gun and shell constitute the system. The internal force developed pushes the shell forward and the gun backward.

Initially the gun and shell are at rest. Therefore, the initial momentum of the system (gun + shell) = 0.



Final momentum of the shell= $\frac{40}{9.81} \times 450 = 1834.86\ kg.m/s$

Final momentum of the gun $= -\frac{6000}{9.81} \times v = -611.62v\ kg.m/s$ (negative because, v is towards left)

Final momentum of the system $= 1834.86 - 611.62v$

Since the forces are internal to the system, the momentum of the system is conserved.

Therefore, initial momentum of the system=final momentum of the system

$$1834.86 - 611.62v = 0$$

$$v = 3\ m/s$$

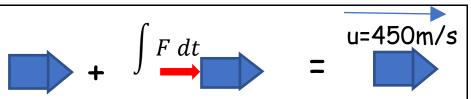
That is, the gun recoils with velocity $v = 3\ m/s$

To find the impulse of the force, let us consider one body constituting the system (say the shell)

Initially, the shell is at rest. Due to the impulse of the force, the shell moves forward with velocity $450\ m/s$.

The initial momentum of the shell = 0

Final momentum of the shell = $\frac{40}{9.81} \times 450 = 1834.86\ kg.m/s$



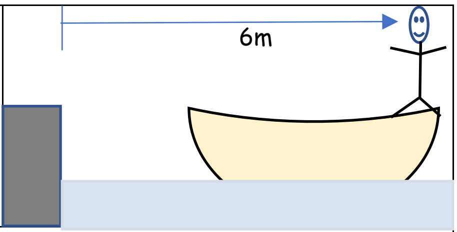
By impulse-momentum principle, Initial momentum + Impulse = Final momentum

$$0 + \int F\,dt = 1834.86 \ggg\gg Impulse = 1834.86\ Ns$$

Average impulsive force $= \frac{1834.86}{0.03} = 61162\ N$

Example 3

A $75\ kg$ man stands in a $100\ kg$ boat so that he is $6m$ from the shore. He walks $3\ m$ in the boat towards shore and stops. How far from the shore will he be at the end of this time? Neglect the friction between the boat and water.



Here, the boat and the man constitute the system.

Initially both man and boat are at rest. Therefore, the initial momentum of the system=0

Assume that the man walks with uniform velocity u towards shore in the boat. He walks $3m$ in time t.

$$u = \frac{3}{t}$$

Momentum of the man when he completes his walk in the boat $= 75 \times u = 75u$

By the time he completes his walk, the boat moves back with him with velocity v

Momentum of the boat+man in the backward direction$= -(75 + 100) \times v$

Net final momentum of the system$= 75 \times \frac{3}{t} - 175v$

Since no external forces are acting on the system and all forces are internal to the system, the momentum of the system is conserved.

$$0 = \frac{225}{t} - 175v$$

$$vt = \frac{225}{175} = 1.28\ m$$

The distance travelled by the boat along with the man backwards$= 1.28\ m$

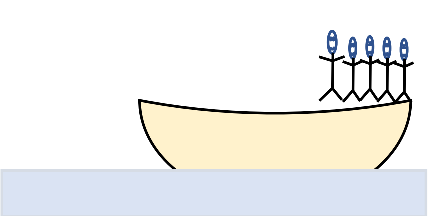
Distance of the man from the shore$= 6 - 3 + 1.28 = 4.28\ m$

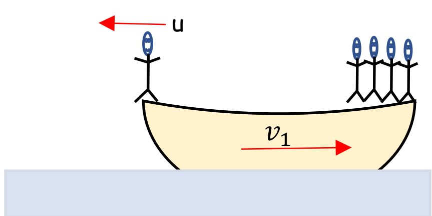
Example 4

Five men lined up at one end of a floating raft, initially at rest, run in succession with velocity $u = 3\ m/s$ relative to the raft and dive off at the far end. Neglecting resistance of the water to horizontal motion of the raft, find its velocity after the last man dives. Each man weighs $w = 760\ N$ and the raft weighs $W = 4450\ N$

Here the system consists of boat + 5 men.

Initially boat+5men are at rest. Initial momentum of the system=0.





The first man runs with velocity u and is about to jump off from the other end. Let the boat moves back with velocity v_1

Momentum of the running man in the forward direction$= \frac{w}{g}u$

Momentum of the boat + 5men in the backward direction$= -\frac{W+5w}{g}v_1$

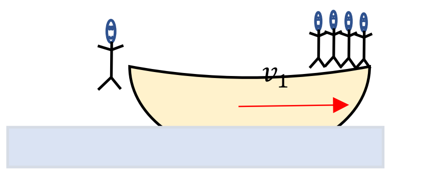
Final momentum of the system$= \frac{w}{g}u - \frac{W+5w}{g}v_1$

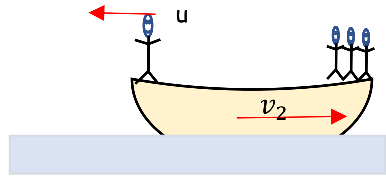
Since no external forces act on the system, the momentum of the system is conserved.

$$0 = \frac{w}{g}u - \frac{W+5w}{g}v_1 \ggg v_1 = \frac{w}{W+5w}u = 0.276$$

Now the first man jumps out.

He is out of the system now. The momentum of boat+4men= $-\frac{W+4w}{g}v_1$





The second man runs and is about to jump off the boat. Now the velocity of the boat backwards be v_2.

Momentum of the running man in the forward direction= $\frac{w}{g}u$

Momentum of the boat + 4 men in the backward direction = $-\frac{W+4w}{g}v_2$

Final momentum of the system= $\frac{w}{g}u - \frac{W+4w}{g}v_2$

By the conservation of momentum,

$$-\frac{W+4w}{g}v_1 = \frac{w}{g}u - \frac{W+4w}{g}v_2$$

$$v_2 - v_1 = \frac{w}{W+4w}u = 0.479$$

Continuing like this, we can write

$$v_3 - v_2 = \frac{w}{W+3w}u = 0.487$$

$$v_4 - v_3 = \frac{w}{W+2w}u = 0.495$$

$$v_5 - v_4 = \frac{w}{W+w}u = 0.504$$

Adding all these, we get

$$v_5 = 2.24\ m/s$$

What will be the velocity of the boat if all the five men run together and jump off???

#Exercise 32

32.1

A wood block weighing $25\ N$ rests on a smooth horizontal surface. A revolver bullet weighing $0.14\ N$ is shot horizontally in to the side of the block. If the block attains a velocity of $3\ m/s$, what is the muzzle velocity of the bullet?

32.2

A jet plane has a mass of $250 \times 10^3\ kg$ and a horizontal velocity of 100 m/s when $t = 0$. If the engines provide a thrust $F = (40 + 0.5t)\ kN$, where t is in seconds, determine the time needed by the plane to attain a velocity of $200\ m/s$. Neglect air resistance and the loss of fuel during the motion.

32.3

A bus has a weight of $75\ kN$ and is travelling to the right at $1.5\ m/s$. Meanwhile a $15\ kN$ car is travelling at $1.2\ m/s$ to the left. If the vehicles crash head-on and become entangled, determine their common velocity just after collision. Also determine the impulse during collision. Assume that the vehicles are free to roll during collision.